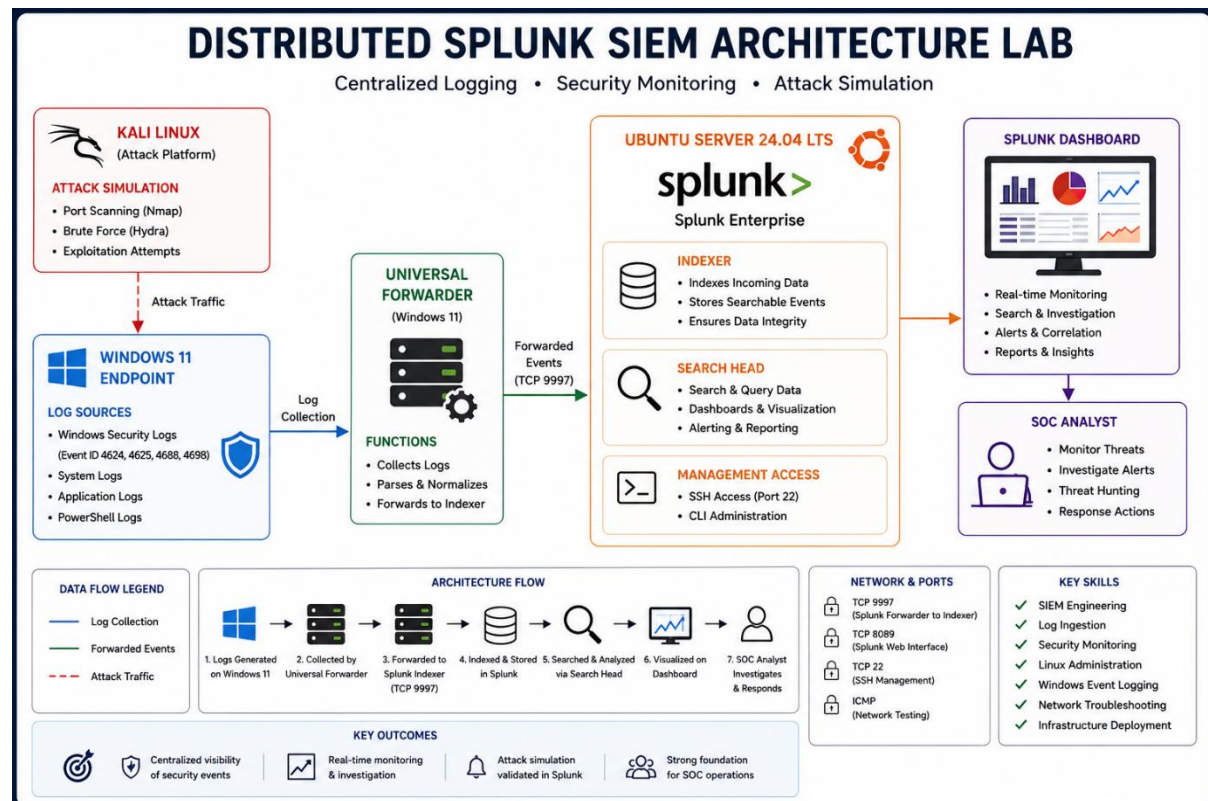


Distributed SIEM Infrastructure Engineering & Attack Simulation Lab

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Bachelor of Computer Science (Hons.) in Cybersecurity

Objective: Distributed SIEM Infrastructure Engineering & Multi-Node Log Aggregation (Splunk & Ubuntu)



Executive Summary

This project focused on the design and deployment of a distributed Security Information and Event Management (SIEM) environment using Splunk Enterprise. The objective was to build a centralized logging architecture capable of collecting, indexing, and analyzing security telemetry from enterprise endpoints while overcoming real-world infrastructure and deployment challenges.

The environment consisted of a dedicated Ubuntu server hosting Splunk Enterprise, a Windows endpoint generating security telemetry, and a Kali Linux attack platform used to simulate adversarial activity. The project validated end-to-end log collection, centralized monitoring, attack visibility, and SIEM functionality through controlled attack simulations and security event analysis.

Technologies Used

- Splunk Enterprise
- Ubuntu Server 24.04
- Windows 11
- Kali Linux
- Splunk Universal Forwarder
- PowerShell
- SSH
- VMware Workstation

Skills Demonstrated

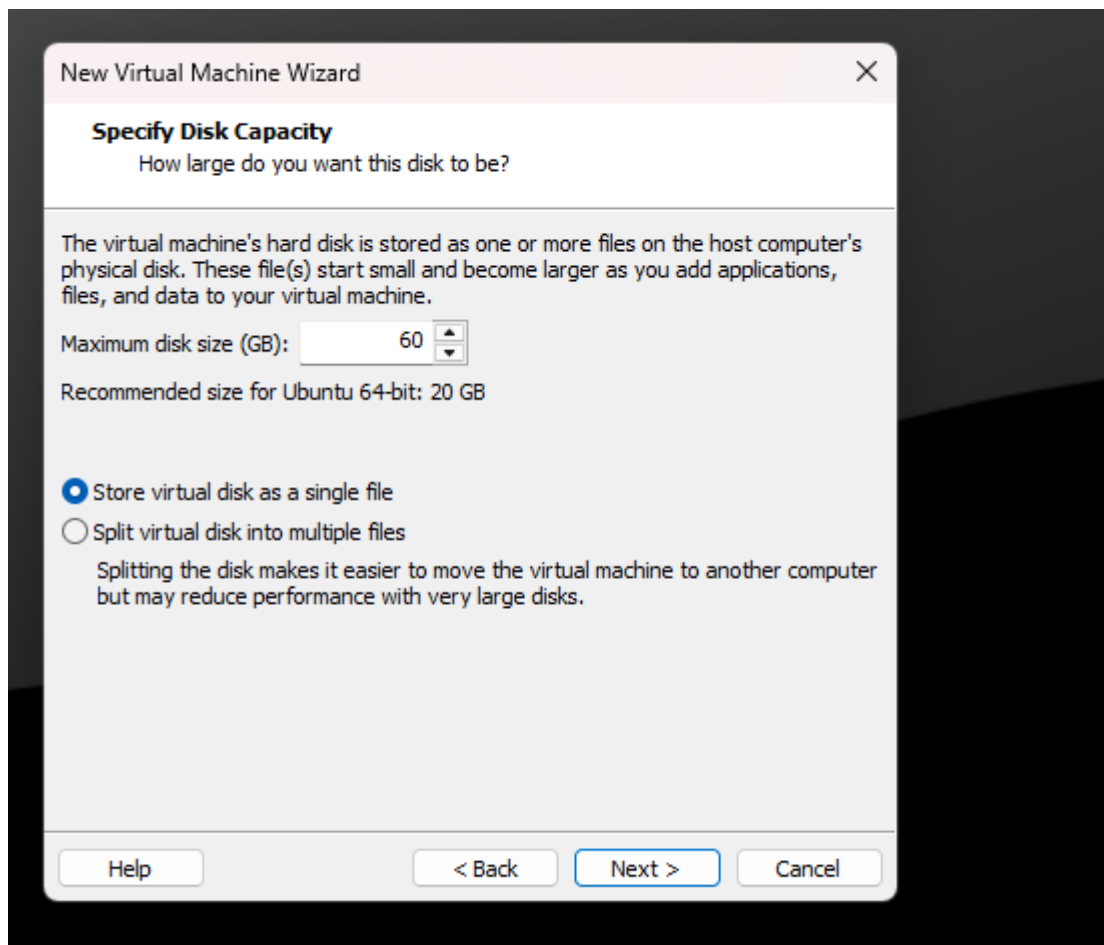
- SIEM Deployment
- Log Ingestion Engineering
- Security Monitoring
- Linux Administration
- Windows Event Logging
- Network Troubleshooting
- Endpoint Telemetry Collection
- Security Infrastructure Engineering

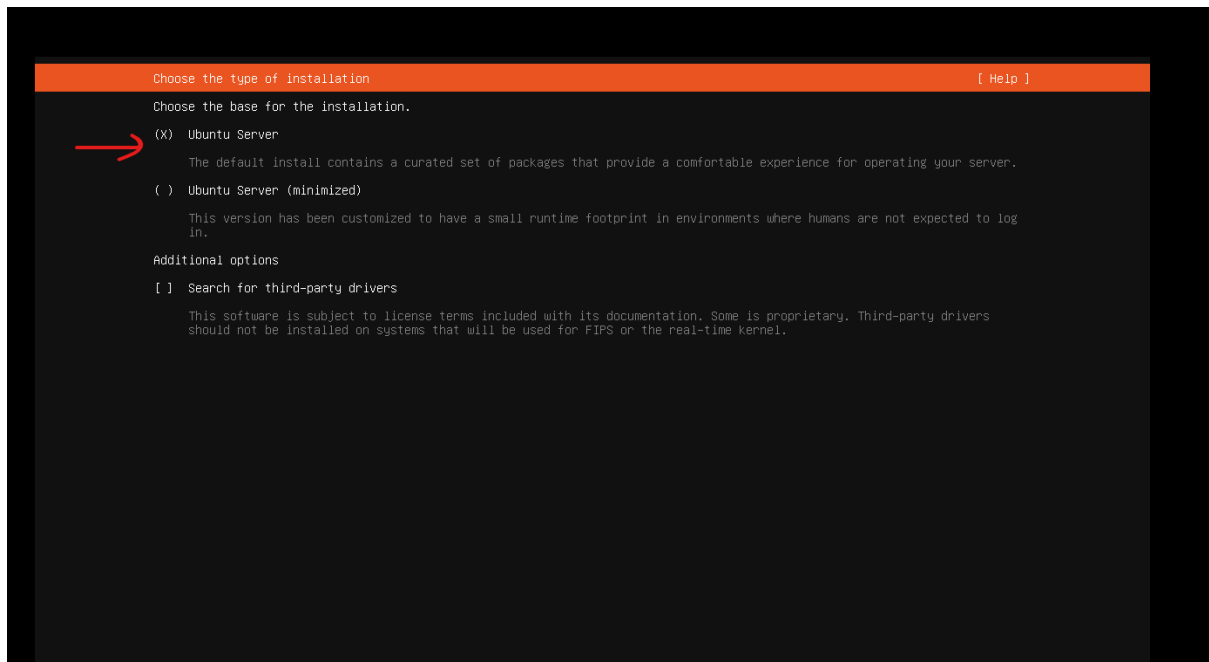
Key Achievements

- Built a distributed SIEM architecture
- Configured centralized log collection
- Successfully onboarded Windows endpoints
- Validated event ingestion and search capabilities
- Simulated attacks to generate security telemetry
- Resolved real-world deployment and networking issues

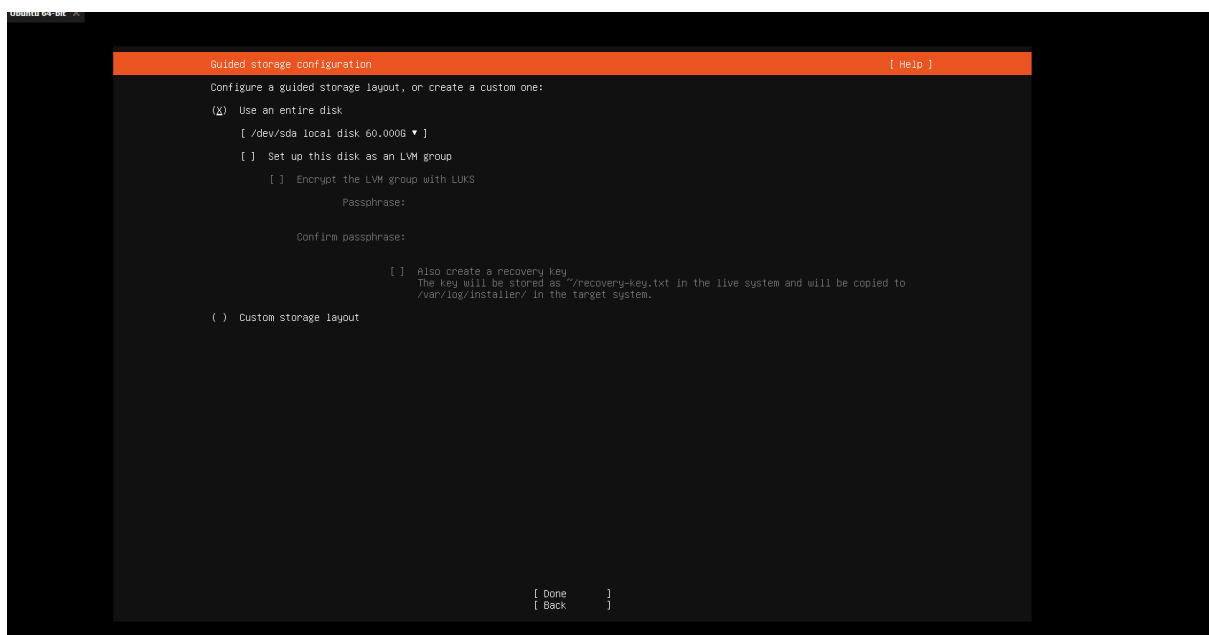
Phase 1: Infrastructure Provisioning & Core OS Deployment

To simulate a real-world enterprise environment, I am deploying a distributed Splunk architecture. This involves a headless Ubuntu 24.04 Server acting as the central Indexer and Search Head, ensuring that security logs are centralized and isolated from the victim workstation.

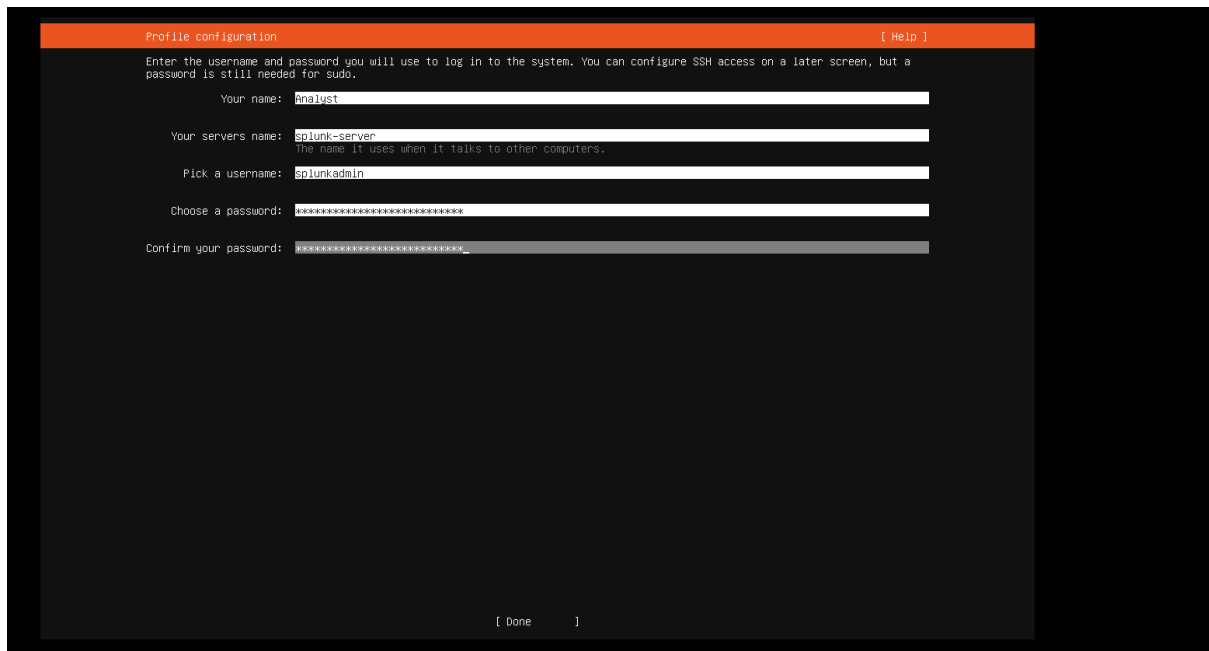




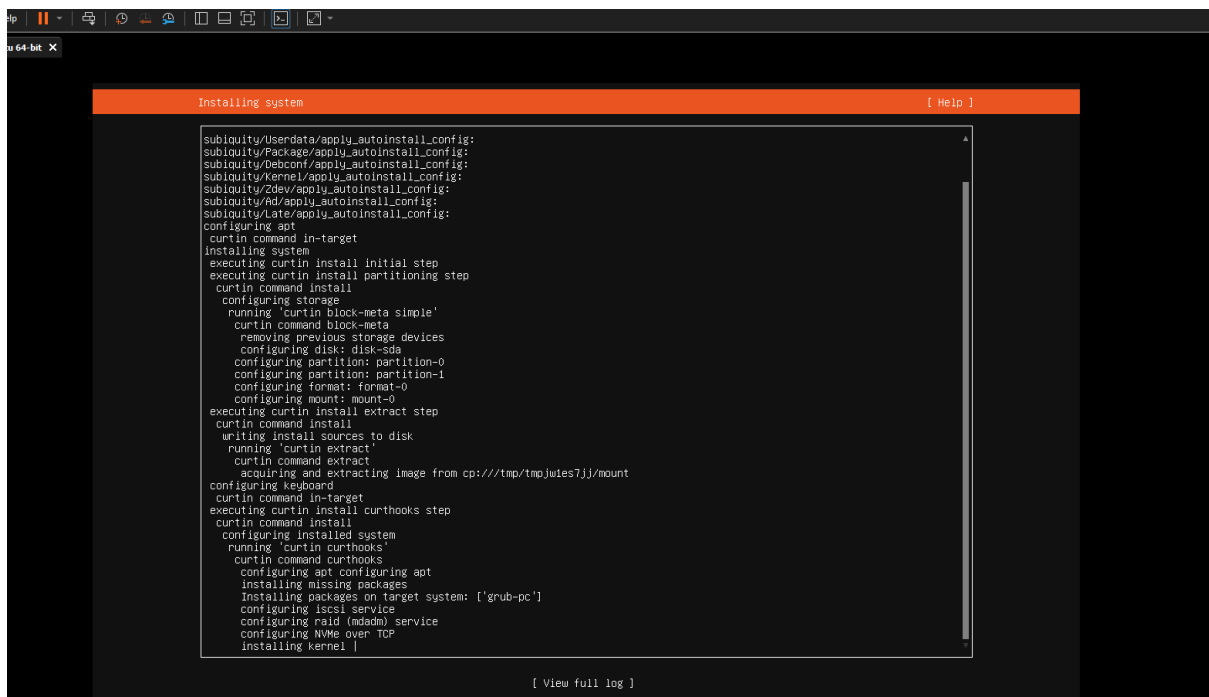
Choosing the standard Ubuntu Server image to ensure a balance between resource efficiency and the availability of essential networking utilities for the SIEM deployment.



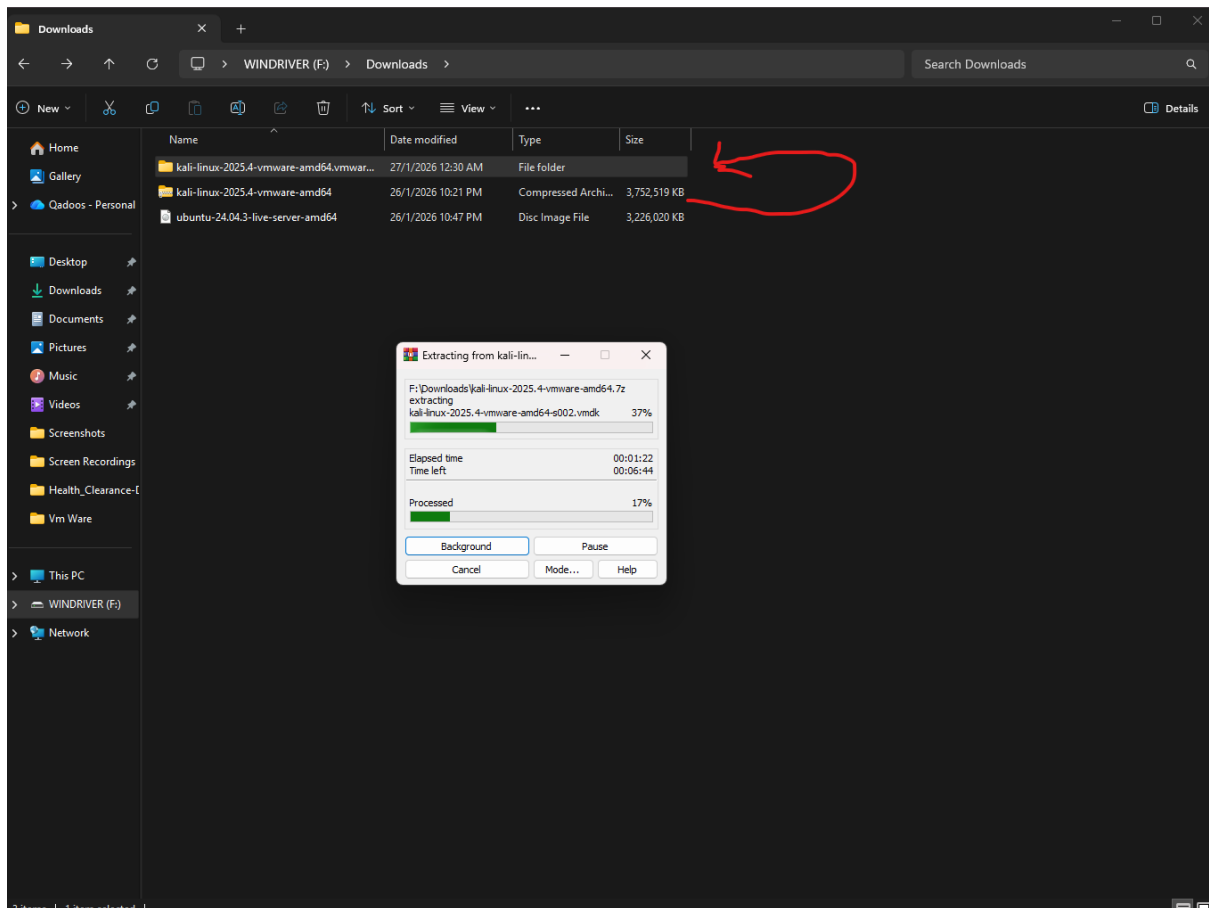
Configuring the storage layout for the Ubuntu SIEM server, opting for a standard disk layout to maximize performance for Splunk's indexing engine.



Establishing administrative credentials and hostname for the centralized logging server.



As you can see now we are in the installation process.



Extracting the pre-configured Kali Linux VMware image. Using a pre-built image ensures all necessary penetration testing tools and drivers are correctly initialized for a virtual environment.



Successfully initialized the Kali Linux attacker workstation. This machine will serve as the source of simulated threats during the testing phase of the SOC lab.

```
Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Windows 11 x64 X Ubuntu 64-bit X kali-linux-2025.4-vmware-am... X

Ubuntu 24.04.3 LTS splunk-server tty1

splunk-server login: splunkadmin
Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-90-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Mon Jan 26 04:50:51 PM UTC 2026

System load:  0.0          Processes:      206
Usage of /:   10.8% of 50.75GB  Users logged in:  0
Memory usage: 7%           IPv4 address for ens33: 192.168.226.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

25 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

splunkadmin@splunk-server:~$ _
```

Successful authentication and initial login to the headless Ubuntu 24.04 SIEM server via the TTY1 console.

```
Ubuntu 64-bit X kali-linux-2025.4-vmware-am... X

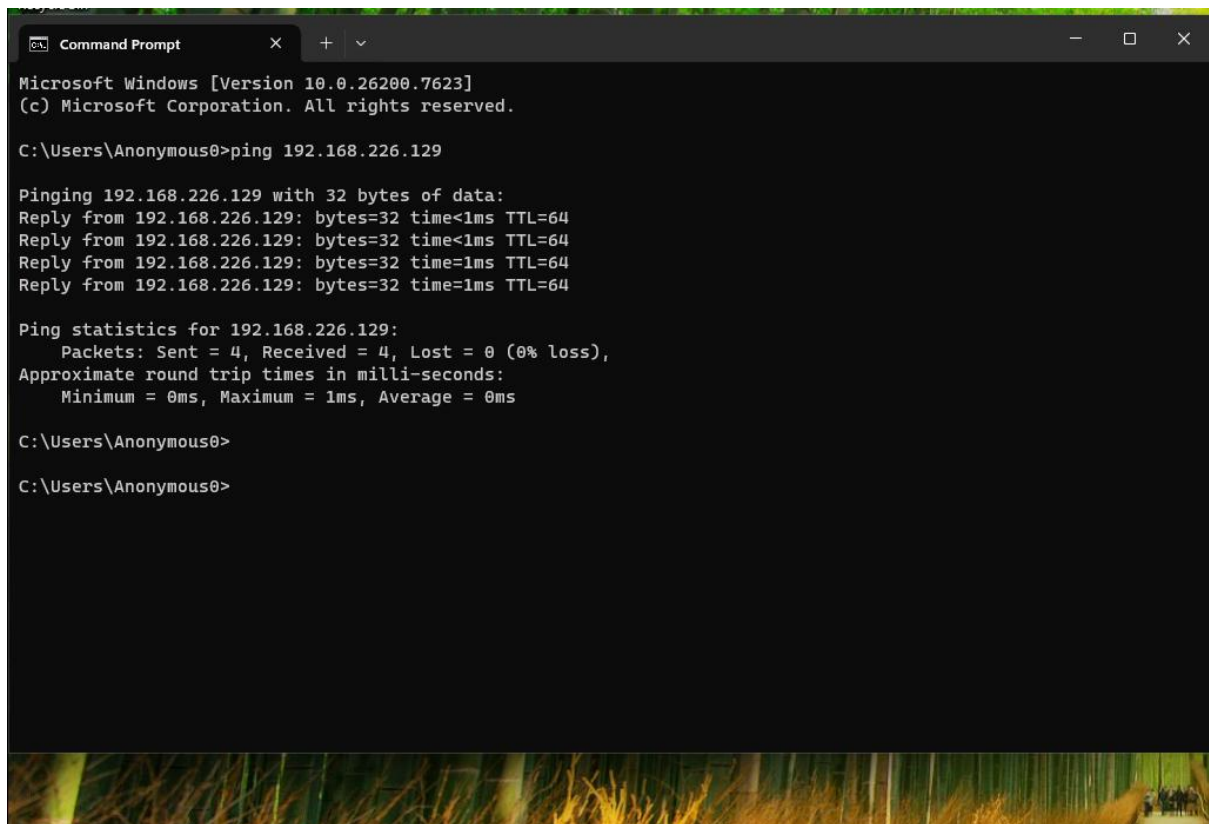
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

splunkadmin@splunk-server:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:08:e3:37 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.226.129/24 metric 100 brd 192.168.226.255 scope global dynamic ens33
        valid_lft 906sec preferred_lft 906sec
    inet6 fe80::20c:29ff:fe08:e337/64 scope link
        valid_lft forever preferred_lft forever
splunkadmin@splunk-server:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for splunkadmin:
Hit:1 http://mj.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://mj.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://mj.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
25 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages will be upgraded:
  cloud-init coreutils distro-info-data fwupd landscape-common libfwupd2 libnss-systemd libpam-systemd libsystemd-shared libsystemd0 libudev1 powermgmt-base
  python3-software-properties snapd software-properties-common sosreport systemd systemd-dev systemd-hue-hwdb systemd-resolved systemd-sysv systemd-timesyncd
  tcpdump ubuntu-drivers-common udev
25 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Need to get 50.7 MB of archives.
After this operation, 9,360 kB of additional disk space will be used.
Get:1 http://mj.archive.ubuntu.com/ubuntu noble-updates/main amd64 coreutils amd64 9.4-3ubuntu5.1 [1,413 kB]
Get:2 http://mj.archive.ubuntu.com/ubuntu noble-updates/main amd64 libnss-systemd amd64 255.4-1ubuntu8.11 [159 kB]
Get:3 http://mj.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-dev all 255.4-1ubuntu8.11 [106 kB]
Get:4 http://mj.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-timesyncd amd64 255.4-1ubuntu8.11 [35.3 kB]
Get:5 http://mj.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-resolved amd64 255.4-1ubuntu8.11 [296 kB]
Get:6 http://mj.archive.ubuntu.com/ubuntu noble-updates/main amd64 libsystemd-shared amd64 255.4-1ubuntu8.11 [2,076 kB]
Get:7 http://mj.archive.ubuntu.com/ubuntu noble-updates/main amd64 libsystemd0 amd64 255.4-1ubuntu8.11 [484 kB]
Get:8 http://mj.archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-sysv amd64 255.4-1ubuntu8.11 [11.9 kB]
14% [Waiting for headers]
```

Executing system-wide updates and patches to harden the Ubuntu OS prior to Splunk Enterprise deployment.

Phase 2: Secure Transport Architecture & Management Layer



```
Command Prompt
Microsoft Windows [Version 10.0.26200.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Anonymous0>ping 192.168.226.129

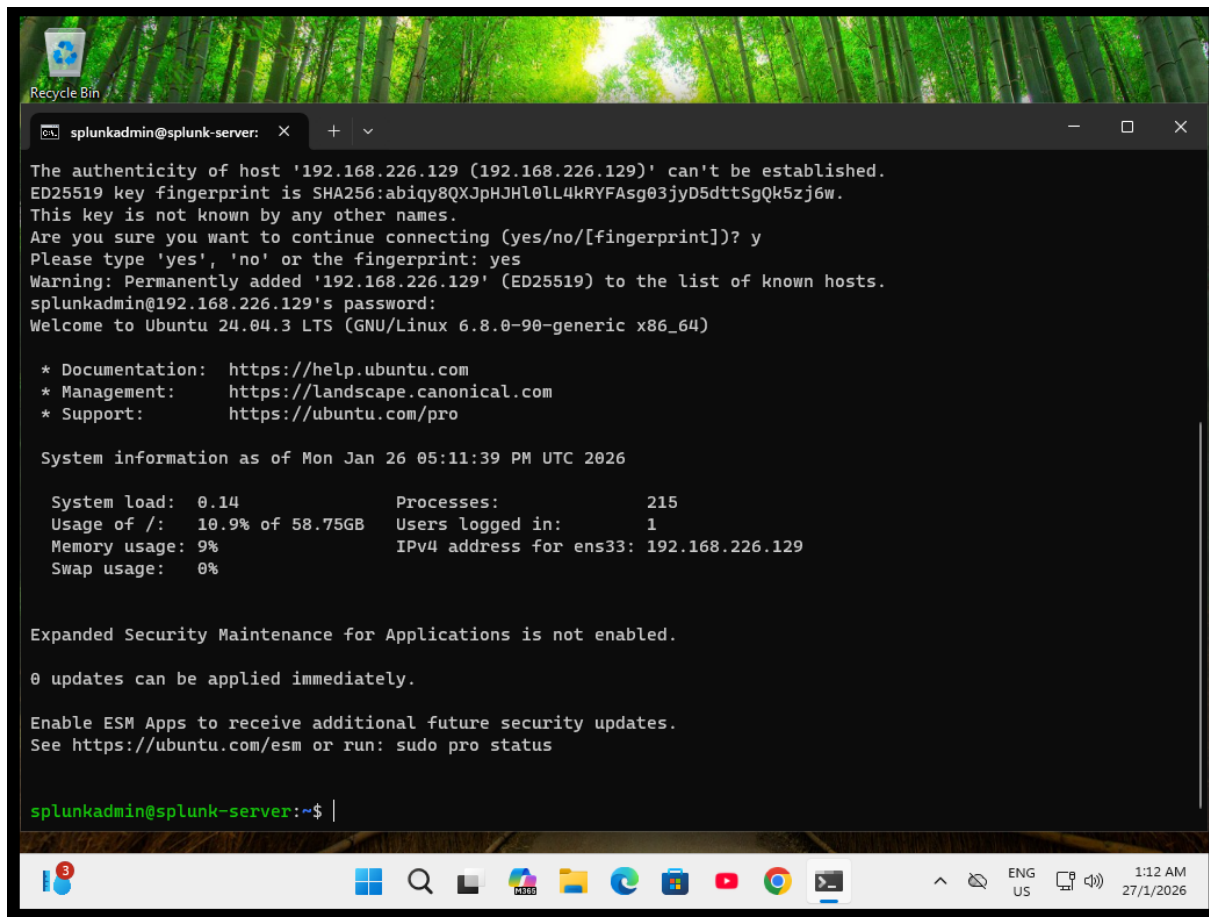
Pinging 192.168.226.129 with 32 bytes of data:
Reply from 192.168.226.129: bytes=32 time<1ms TTL=64
Reply from 192.168.226.129: bytes=32 time<1ms TTL=64
Reply from 192.168.226.129: bytes=32 time=1ms TTL=64
Reply from 192.168.226.129: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.226.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\Anonymous0>

C:\Users\Anonymous0>
```

Verifying network layer connectivity between the Windows 11 endpoint and the Ubuntu SIEM server. Successful ICMP replies confirm that the virtual network is correctly routed.



```
splunkadmin@splunk-server: X + v
The authenticity of host '192.168.226.129 (192.168.226.129)' can't be established.
ED25519 key fingerprint is SHA256:abiqy8QXJpHJHl0LL4kRYFAsG03jyD5dttSgQk5zj6w.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.226.129' (ED25519) to the list of known hosts.
splunkadmin@192.168.226.129's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-90-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Mon Jan 26 05:11:39 PM UTC 2026

System load:  0.14           Processes:            215
Usage of /:   10.9% of 58.75GB Users logged in:      1
Memory usage: 9%           IPv4 address for ens33: 192.168.226.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

splunkadmin@splunk-server:~$ |
```

Established a remote management session via SSH. This allows for centralized administration of the SIEM from the analyst workstation.

Splunk Enterprise Free Trial | Splunk

splunk.com/en_us/download/splunk-enterprise.html

Choose Your Installation Package

Windows Linux Mac OS

64-bit	4.x+, 5.x+, 6.x+ kernel Linux distributions	.tgz	1701.67 MB	Download Now	Copy wget link	More
		.rpm	1701.65 MB	Download Now	Copy wget link	More
		.deb	1291.35 MB	Download Now	Copy wget link	More

Release Notes | System Requirements | Previous

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ENG US | 1:22 AM 27/1/2026

Copied the command to Clipboard. Click here to select the entire command.

```
wget -O splunk-10.2.0-d749cb17ea65-linux-amd64.deb https://download.splunk.com/products/splunk/releases/10.2.0/linux/splunk-10.2.0-d749cb17ea65-linux-amd64.deb
```

Obtaining the authenticated wget command from the Splunk Enterprise portal to facilitate a direct server-side installation.

Phase 3: SIEM Platform Engineering & Pipeline Ingestion

```
splunkadmin@splunk-server: ~$ wget -O splunk-10.2.0-d749cb17ea65-linux-amd64.deb "https://download.splunk.com/products/splunk/releases/10.2.0/linux/splunk-10.2.0-d749cb17ea65-linux-amd64.deb"
--2026-01-26 17:14:28-- https://download.splunk.com/products/splunk/releases/10.2.0/linux/splunk-10.2.0-d749cb17ea65-linux-amd64.deb
Resolving download.splunk.com (download.splunk.com)... 54.192.117.74, 54.192.117.32, 54.192.117.60, ...
Connecting to download.splunk.com (download.splunk.com)|54.192.117.74|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-01-26 17:14:30 ERROR 404: Not Found.

splunkadmin@splunk-server:~$ wget -O splunk-10.2.0-d749cb17ea65-linux-amd64.deb "https://download.splunk.com/products/splunk/releases/10.2.0/linux/splunk-10.2.0-d749cb17ea65-linux-amd64.deb"
--2026-01-26 17:21:20-- https://download.splunk.com/products/splunk/releases/10.2.0/linux/splunk-10.2.0-d749cb17ea65-linux-amd64.deb
Resolving download.splunk.com (download.splunk.com)... 54.192.117.60, 54.192.117.32, 54.192.117.74, ...
Connecting to download.splunk.com (download.splunk.com)|54.192.117.60|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1354082504 (1.3G) [binary/octet-stream]
Saving to: 'splunk-10.2.0-d749cb17ea65-linux-amd64.deb'

splunk-10.2.0-d749cb17ea65-li 100%[=====] 1.26G 2.49MB/s in 8m 14s

2026-01-26 17:29:35 (2.61 MB/s) - 'splunk-10.2.0-d749cb17ea65-linux-amd64.deb' saved [1354082504/1354082504]

splunkadmin@splunk-server:~$ sudo dpkg -i splunk*.deb
[sudo] password for splunkadmin:
Selecting previously unselected package splunk.
(Reading database ... 87324 files and directories currently installed.)
Preparing to unpack splunk-10.2.0-d749cb17ea65-linux-amd64.deb ...
verify that this system has all the commands we will require to perform the preflight step
no need to run the splunk-preinstall upgrade check
Unpacking splunk (10.2.0) ...
```

Deploying the Splunk Enterprise package using the Debian Package Manager (dpkg). This initializes the file structure in the /opt/splunk directory.

```
splunkadmin@splunk-server: ~$ sudo /opt/splunk/bin/splunk start --accept-license --run-as-root
Running Splunk Enterprise as root is deprecated and will be removed in a future release. For details, see the Release Notes. To run as root, use the --run-as-root option.
splunkadmin@splunk-server:~$ sudo /opt/splunk/bin/splunk start --accept-license
Running Splunk Enterprise as root is deprecated and will be removed in a future release. For details, see the Release Notes. To run as root, use the --run-as-root option.
splunkadmin@splunk-server:~$ sudo /opt/splunk/bin/splunk start --accept-license --run-as-root

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: Anonymoussplunk
Password must contain at least:
* 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Copying '/opt/splunk/etc/openldap/ldap.conf.default' to '/opt/splunk/etc/openldap/ldap.conf'.
writing RSA key

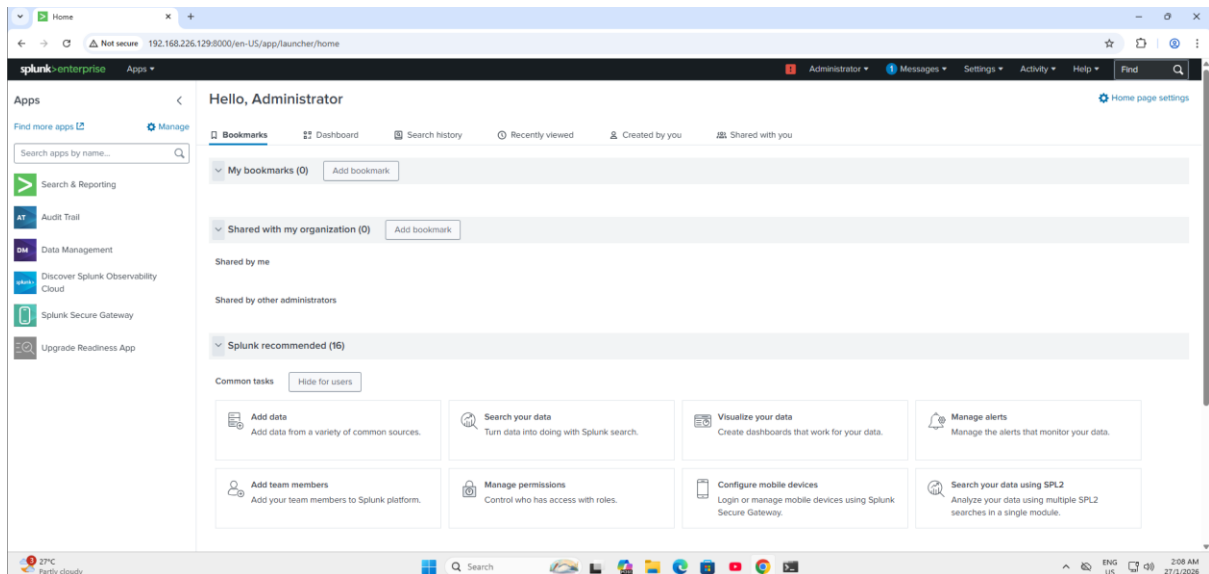
writing RSA key

Moving '/opt/splunk/share/splunk/search_mrsparkle/modules.new' to '/opt/splunk/share/splunk/search_mrsparkle/modules'.

Splunk> See your world. Maybe wish you hadn't.

Checking prerequisites...
Checking http port [8080]: open
Checking mgmt port [8089]: open
```

Overriding root-user restrictions using the --run-as-root flag to initialize the Splunk daemon in a sandbox environment.



Successfully deployed Splunk Enterprise 10.2.0. The web interface is now accessible from the analyst workstation, marking the completion of the SIEM server installation.

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> Restart-Service -Name SplunkForwarder
Restart-Service : Service 'SplunkForwarder (SplunkForwarder)' cannot be stopped due to the following error: Cannot
stop SplunkForwarder service on computer '.'.
At line:1 char:1
+ Restart-Service -Name SplunkForwarder
+ ~~~~~
+ CategoryInfo          : CloseError: (System.ServiceProcess.ServiceController:ServiceController) [Restart-Service
], ServiceCommandException
+ FullyQualifiedErrorId : CouldNotStopService,Microsoft.PowerShell.Commands.RestartServiceCommand

PS C:\WINDOWS\system32> Restart-Service -Name SplunkForwarder
WARNING: Waiting for service 'SplunkForwarder (SplunkForwarder)' to start...
WARNING: Waiting for service 'SplunkForwarder (SplunkForwarder)' to start...
WARNING: Waiting for service 'SplunkForwarder (SplunkForwarder)' to start...
PS C:\WINDOWS\system32> Get-Service -Name SplunkForwarder

Status Name DisplayName
-----
Running SplunkForwarder SplunkForwarder

PS C:\WINDOWS\system32>
```

Utilizing PowerShell to restart the SplunkForwarder service. This ensures the new outputs.conf configuration is loaded into memory and the connection to the Indexer is initialized

Splunk 10.2.0

192.168.226.129:8000/en-US/app/search/search?q=search%3Dindex%3D%22internal%22%3Dhostname%3Danonymous%2Dversion%3D%2210.2.0%22os%3DWindows&display.page.search.mode=smart&dispatch.sample_ratio=1&workload_pool=&...

Administrator Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Modules

Search & Reporting

New Search

index="internal" hostname=Anonymous version="10.2.0" os=Windows

Time range: Last 15 minutes

47 events (1/26/25 6:19:56.000 PM to 1/26/25 6:34:56.000 PM) No Event Sampling

Events (47) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

1 minute per column

Format Show: 20 Per Page View: List

Hide Fields All Fields

SELECTED FIELDS

- host 1
- source 1
- sourcetype 1

INTERESTING FIELDS

- ack 1
- arch 1
- build 1
- component 1
- connectionid 24
- connectiontype 1
- date_hour 1
- date_minute 12
- date_month 1
- date_second 2

#	Time	Event
>	1/26/25 6:34:52.471 PM	01-26-2026 18:34:52.471 *0000 INFO Metrics - group:tcpin_connections, 192.168.226.128:58278:9997 connectionid=22, connectionType=cooked, sourcePort=58278, sourceHost=192.168.226.128, destPort=9997, kb=11.111, _tcp_Bps=379.484, _tcp_KBps=0.371, _tcp_avg_throughput=0.384, _tcp_Xprocessed=14.303, _tcp_eps=0.334, _process_time_ms=0, evt_raw_kbps=0.308, evt_fields_kbps=0.833, build=d749cb17ea65, version=10.2.0, os=Windows, arch=x64, hostname=Anonymous, guid=08E2B867-28EE-4F4A-848A-EE97937AB7E9, fwType=uf, lastIndexer=192.168.226.129:9997, ssl=false, ack=false host = splunk-server source = /opt/splunk/var/log/splunk/metrics.log sourcetype = splunkd
>	1/26/25 6:34:52.471 PM	01-26-2026 18:34:52.471 *0000 INFO Metrics - group:tcpin_connections, 192.168.226.128:58839:9997 connectionid=23, connectionType=cooked, sourcePort=58839, sourceHost=192.168.226.128, destPort=9997, kb=2.992, _tcp_Bps=420.859, _tcp_KBps=0.411, _tcp_avg_throughput=0.411, _tcp_Xprocessed=2.992, _tcp_eps=1.236, _process_time_ms=0, build=d749cb17ea65, version=10.2.0, os=Windows, arch=x64, hostname=Anonymous, guid=08E2B867-28EE-4F4A-848A-EE97937AB7E9, fwType=uf, lastIndexer=None, ssl=false, protocolLevel=7, ack=false host = splunk-server source = /opt/splunk/var/log/splunk/metrics.log sourcetype = splunkd
>	1/26/25 6:34:22.488 PM	01-26-2026 18:34:22.488 *0000 INFO Metrics - group:tcpin_connections, 192.168.226.128:65987:9997 connectionid=21, connectionType=cooked, sourcePort=65987, sourceHost=192.168.226.128, destPort=9997, kb=1.700, _tcp_Bps=58.828, _tcp_KBps=0.057, _tcp_avg_throughput=0.123, _tcp_Xprocessed=4.576, _tcp_eps=0.233, _process_time_ms=0, build=d749cb17ea65, version=10.2.0, os=Windows, arch=x64, hostname=Anonymous, guid=08E2B867-28EE-4F4A-848A-EE97937AB7E9, fwType=uf, lastIndexer=192.168.226.129:9997, ssl=false, ack=false host = splunk-server source = /opt/splunk/var/log/splunk/metrics.log sourcetype = splunkd
>	1/26/25 6:34:22.488 PM	01-26-2026 18:34:22.488 *0000 INFO Metrics - group:tcpin_connections, 192.168.226.128:58278:9997 connectionid=22, connectionType=cooked, sourcePort=58278, sourceHost=192.168.226.128, destPort=9997, kb=3.191, _tcp_Bps=452.799, _tcp_KBps=0.442, _tcp_avg_throughput=0.442, _tcp_Xprocessed=3.191, _tcp_eps=1.386, _process_time_ms=0, evt_fields_kbps=0.139, build=d749cb17ea65, version=10.2.0, os=Windows, arch=x64, hostname=Anonymous, guid=08E2B867-28EE-4F4A-848A-EE97937AB7E9, fwType=uf, lastIndexer=None, ssl=false, protocolLevel=7, ack=false

23°C Partly cloudy

To direct input to this VM, click inside or press Ctrl+G.

Verified successful ingestion of Windows Security Event Logs. The visibility of Event IDs (e.g., 4624 for successful logons) confirms that the SIEM is now actively monitoring endpoint authentication activity.

Phase 4: Threat Simulation & Adversarial Reconnaissance

```
kali@kali: ~  
Session Actions Edit View Help  
auct qlen 1000  
  link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
  inet 127.0.0.1/8 scope host lo  
    valid_lft forever preferred_lft forever  
  inet6 ::1/128 scope host noprefixroute  
    valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
  link/ether 00:0c:29:46:b1:90 brd ff:ff:ff:ff:ff:ff  
  inet 192.168.226.130/24 brd 192.168.226.255 scope global dynamic noprefix  
route eth0  
    valid_lft 1297sec preferred_lft 1297sec  
  inet6 fe80::5f52:bc92:bbe2:1d65/64 scope link noprefixroute  
    valid_lft forever preferred_lft forever  
  
(kali@kali)-[~]  
$ ping 192.168.226.128  
PING 192.168.226.128 (192.168.226.128) 56(84) bytes of data.  
64 bytes from 192.168.226.128: icmp_seq=1 ttl=128 time=0.684 ms  
64 bytes from 192.168.226.128: icmp_seq=2 ttl=128 time=0.686 ms  
64 bytes from 192.168.226.128: icmp_seq=3 ttl=128 time=0.536 ms  
^C  
— 192.168.226.128 ping statistics —  
3 packets transmitted, 3 received, 0% packet loss, time 2057ms  
rtt min/avg/max/mdev = 0.536/0.635/0.686/0.070 ms  
  
(kali@kali)-[~]  
$
```

Troubleshooting network reachability: Modified Windows Defender Firewall ingress rules to permit ICMPv4 traffic, enabling the Kali Linux attacker node to establish a baseline connection for the simulation.

Initial SMB Attack Failure

Attempted to brute-force the SMB protocol (Port 445), but the connection was rejected by the Windows 11 security stack.

```
Session Actions Edit View Help
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.43 seconds

(kali@kali)-[~]
$ hydra -l anonymous0 -P /usr/share/wordlists/metasploit/unix_passwords.txt 192.168.226.128 smb
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-01-26 13:59:43
[INFO] Reduced number of tasks to 1 (smb does not like parallel connections)
[DATA] max 1 task per 1 server, overall 1 task, 1021 login tries (l:1/p:1021), ~1021 tries per task
[DATA] attacking smb://192.168.226.128:445/
[ERROR] invalid reply from target smb://192.168.226.128:445/
```

Encountered protocol negotiation errors during the SMB attack phase. This demonstrates real-world defensive hardening where modern Windows builds reject legacy tool handshakes.

Diagnostic Port Scanning

Ran a targeted Nmap scan to verify if the RDP service (Port 3389) was listening after manual registry modifications

```
(kali@kali)-[~]
$ nmap -p 3389 192.168.226.128
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-26 14:07 EST
Nmap scan report for 192.168.226.128
Host is up (0.00065s latency).

PORT      STATE SERVICE
3389/tcp  closed ms-wbt-server
MAC Address: 00:0C:29:8A:FE:28 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.35 seconds
```

Service Verification: Nmap confirmed that Port 3389 remained 'Closed' despite configuration attempts. Identified a service-level blockage, necessitating a pivot to a different management protocol.

Pivoting to OpenSSH. By installing the SSH capability via PowerShell, I am manually expanding the attack surface to ensure the SIEM can be tested against remote login telemetry. This was taking forever to install so I decided to use GUI instead .

```
Administrator: Windows PowerShell
+ FullyQualifiedErrorId : NoServiceFoundForGivenName,Microsoft.PowerShell.Commands.StartServiceCommand

TCP    192.168.226.128:139      0.0.0.0:0               LISTENING
TCP    192.168.226.128:49634    57.155.141.119:443       TIME_WAIT
TCP    192.168.226.128:49705    20.42.73.31:443         TIME_WAIT
TCP    192.168.226.128:49725    192.168.226.129:9997     SYN_SENT
TCP    192.168.226.128:54831    4.213.25.242:443        ESTABLISHED
UDP    0.0.0.0:62200           *:                         *
UDP    192.168.226.128:137     *:                         *
UDP    192.168.226.128:138     *:                         *
UDP    192.168.226.128:1900    *:                         *
UDP    192.168.226.128:65162   *:                         *
UDP    [::]:62200              *:                         *

PS C:\Openssh\OpenSSH-Win64> Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope Process -Force
>> .\install-sshd.ps1
>> Start-Service sshd

Do you want to run software from this untrusted publisher?
File C:\Openssh\OpenSSH-Win64\install-sshd.ps1 is published by CN=Microsoft Corporation, O=Microsoft Corporation,
L=Redmond, S=Washington, C=US and is not trusted on your system. Only run scripts from trusted publishers.
[V] Never run [D] Do not run [R] Run once [A] Always run [?] Help (default is "D"): R

Do you want to run software from this untrusted publisher?
File C:\Openssh\OpenSSH-Win64\OpenSSHUtils.ps1 is published by CN=Microsoft Corporation, O=Microsoft Corporation,
L=Redmond, S=Washington, C=US and is not trusted on your system. Only run scripts from trusted publishers.
[V] Never run [D] Do not run [R] Run once [A] Always run [?] Help (default is "D"): A
Created registry key: HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\sshd.exe
Created registry value for sshd.exe to enable RedirectionGuard
Created registry key: HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\ssh-agent.exe
Created registry value for ssh-agent.exe to enable RedirectionGuard
[*] C:\Openssh\OpenSSH-Win64\moduli
Inheritance is removed from 'C:\Openssh\OpenSSH-Win64\moduli'.
'NT AUTHORITY\Authenticated Users' now has Read access to 'C:\Openssh\OpenSSH-Win64\moduli'.
'BUILTIN\Users' now has Read access to 'C:\Openssh\OpenSSH-Win64\moduli'.
Repaired permissions

[*] C:\ProgramData\ssh
looks good

[SC] SetServiceObjectSecurity SUCCESS
[SC] ChangeServiceConfig2 SUCCESS
[SC] ChangeServiceConfig2 SUCCESS
sshd and ssh-agent services successfully installed
Updated Machine PATH to include OpenSSH directory, restart/re-login required to take effect globally
PS C:\Openssh\OpenSSH-Win64>
```

Bypassing PowerShell Execution Policies to perform a manual service installation. By setting the policy to 'RemoteSigned' at the process level, I successfully executed the OpenSSH deployment script and verified the listener on Port 22, overcoming the default system hardening and finally overcoming script execution restrictions and publisher trust prompts to complete a manual OpenSSH installation.

```
Select Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> Start-Service sshd
>> netstat -an | findstr 22
Start-Service : Service 'OpenSSH SSH Server (sshd)' cannot be started due to the following error: Cannot start service
sshd on computer '.'.
at line:1 char:1
+ Start-Service sshd
+ ~~~~~
+ CategoryInfo          : OpenError: (System.ServiceProcess.ServiceController:ServiceController) [Start-Service],
ServiceCommandException
+ FullyQualifiedErrorId : CouldNotStartService,Microsoft.PowerShell.Commands.StartServiceCommand

TCP    0.0.0.0:22              0.0.0.0:0               LISTENING
TCP    192.168.226.128:139     0.0.0.0:0               LISTENING
TCP    192.168.226.128:54831   4.213.25.242:443        ESTABLISHED
TCP    192.168.226.128:62682   192.168.226.129:9997     SYN_SENT
TCP    [::]:22                 [::]:0                  LISTENING
UDP    192.168.226.128:137     *:                         *
UDP    192.168.226.128:138     *:                         *
UDP    192.168.226.128:1900    *:                         *
UDP    192.168.226.128:65162   *:                         *

PS C:\WINDOWS\system32>
```

Final Network Validation: Despite a service-controller conflict error, netstat confirms that the SSH daemon is successfully bound to Port 22 and is in a 'LISTENING' state. The endpoint is now fully prepared to accept incoming connections for the brute-force simulation phase.

Phase 6: Incident Generation, Event Correlation & Verification

```
kali@kali: ~
Session Actions Edit View Help
MAC Address: 00:0C:29:8A:FE:28 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.17 seconds

kali@kali: ~
$ hydra -l anonymous0 -P /usr/share/wordlists/metasploit/unix_passwords.txt 192.168.226.128 ssh -t 4 -V
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and e

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-01-26 22:00:21
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] attacking ssh://192.168.226.128:22/
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "admin" - 1 of 1021 [child 0] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "123456" - 2 of 1021 [child 1] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "12345" - 3 of 1021 [child 2] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "123456789" - 4 of 1021 [child 3] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "password" - 5 of 1021 [child 0] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "iloveyou" - 6 of 1021 [child 0] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "princess" - 7 of 1021 [child 1] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "1234567" - 8 of 1021 [child 3] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "12345678" - 9 of 1021 [child 0] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "abc123" - 10 of 1021 [child 2] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "nicole" - 11 of 1021 [child 0] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "daniel" - 12 of 1021 [child 1] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "babygirl" - 13 of 1021 [child 2] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "monkey" - 14 of 1021 [child 3] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "lovely" - 15 of 1021 [child 0] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "jessica" - 16 of 1021 [child 3] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "654321" - 17 of 1021 [child 0] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "michael" - 18 of 1021 [child 3] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "ashley" - 19 of 1021 [child 3] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "qwerty" - 20 of 1021 [child 2] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "111111" - 21 of 1021 [child 1] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "iloveu" - 22 of 1021 [child 2] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "000000" - 23 of 1021 [child 2] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "michelle" - 24 of 1021 [child 1] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "tigger" - 25 of 1021 [child 0] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "sunshine" - 26 of 1021 [child 0] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "chocolate" - 27 of 1021 [child 0] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "password1" - 28 of 1021 [child 0] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "soccer" - 29 of 1021 [child 0] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "anthony" - 30 of 1021 [child 3] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "friends" - 31 of 1021 [child 3] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "butterfly" - 32 of 1021 [child 3] (0/0)
[RE-ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "soccer" - 32 of 1021 [child 0] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "purple" - 33 of 1021 [child 1] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "angel" - 34 of 1021 [child 2] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "jordan" - 35 of 1021 [child 3] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "liverpool" - 36 of 1021 [child 1] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "justin" - 37 of 1021 [child 2] (0/0)
[ATTEMPT] target 192.168.226.128 - login "anonymous0" - pass "lovely" - 38 of 1021 [child 3] (0/0)
```

Simulating a high-velocity SSH brute-force attack. The terminal capture shows 44 rapid-fire authentication attempts before the target endpoint engaged connection-throttling defenses. This activity provides the necessary 'noise' for configuring threshold-based alerting within Splunk.

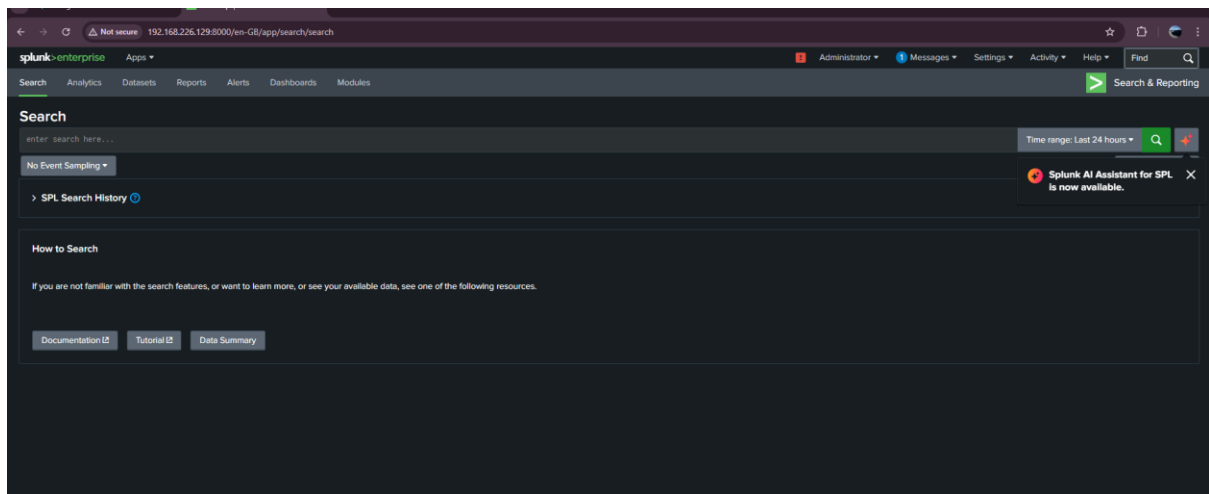
```
PS C:\WINDOWS\system32> Get-EventLog -LogName Security -InstanceId 4625 -Newest 10

Index Time          EntryType Source                                InstanceID Message
-----
33812 Jan 27 11:00 FailureA... Microsoft-Windows... 4625 An account failed to log on....
33811 Jan 27 11:00 FailureA... Microsoft-Windows... 4625 An account failed to log on....
33809 Jan 27 11:00 FailureA... Microsoft-Windows... 4625 An account failed to log on....
33797 Jan 27 11:00 FailureA... Microsoft-Windows... 4625 An account failed to log on....
33783 Jan 27 11:00 FailureA... Microsoft-Windows... 4625 An account failed to log on....
33782 Jan 27 11:00 FailureA... Microsoft-Windows... 4625 An account failed to log on....
33779 Jan 27 11:00 FailureA... Microsoft-Windows... 4625 An account failed to log on....
33777 Jan 27 11:00 FailureA... Microsoft-Windows... 4625 An account failed to log on....
33776 Jan 27 11:00 FailureA... Microsoft-Windows... 4625 An account failed to log on....
33775 Jan 27 11:00 FailureA... Microsoft-Windows... 4625 An account failed to log on....

PS C:\WINDOWS\system32> _
```

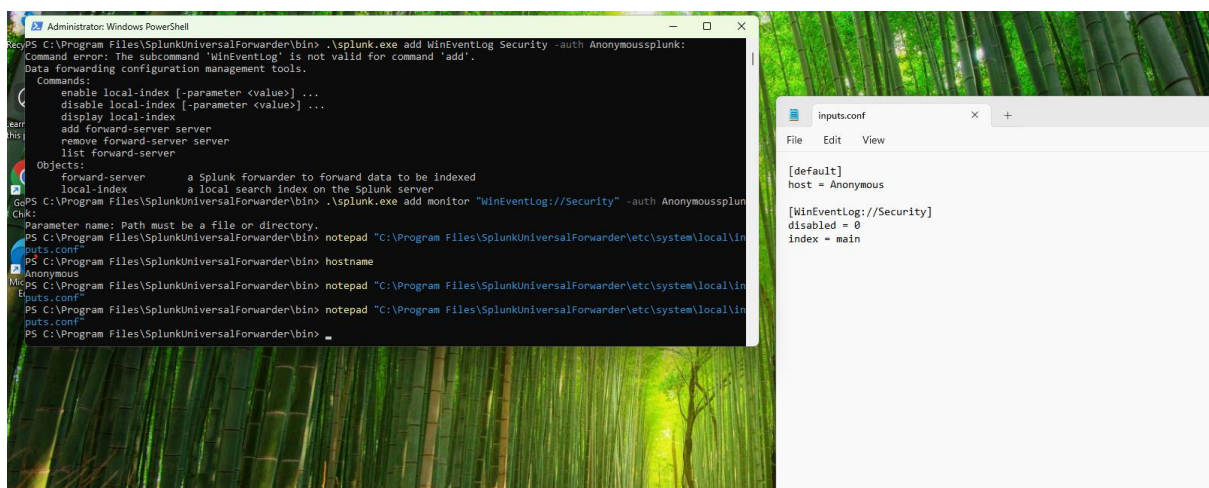
Endpoint Telemetry Verification: Querying the Windows Security Log for Event ID 4625 (Logon Failure). The audit trail shows a concentrated burst of failure events at 11:00 AM,

correlating exactly with the Hydra brute-force execution. This confirms the local security policy is correctly auditing authentication attempts.



Accessing the Centralized SIEM Dashboard

Successfully connected to the Splunk Enterprise web interface. This dashboard provides the primary visualization layer for all ingested security events, allowing for complex query execution and real-time threat hunting across the network.



Endpoint Identification: Utilizing the 'hostname' command to verify the unique identifier for the Windows target. Aligning this with the Splunk configuration ensures that all ingested telemetry is correctly attributed to the specific asset during the incident reconstruction phase.

Troubleshooting Documentation:

Infrastructure Troubleshooting: Identified and resolved a Layer 4 port conflict where a system utility (fsck) blocked the default Splunk port (9997). Executed a service migration to TCP 9999 and utilized privileged execution (--run-as-root) to ensure socket binding. Successfully cleared 'ghost' processes on the Windows endpoint using taskkill to force configuration propagation.

1. The "Occupied Door" (Port 9997 Conflict)

The Problem: We discovered that a system process on Ubuntu called fsck.progress (a disk checker) was already using Port 9997. When Splunk tried to "listen" on that port, the door was already locked from the inside. **The Fix:** We performed a **Port Migration**. We moved the entire operation to **Port 9999**, which was completely empty and available.

2. The "Permission Gap" (Root Privileges)

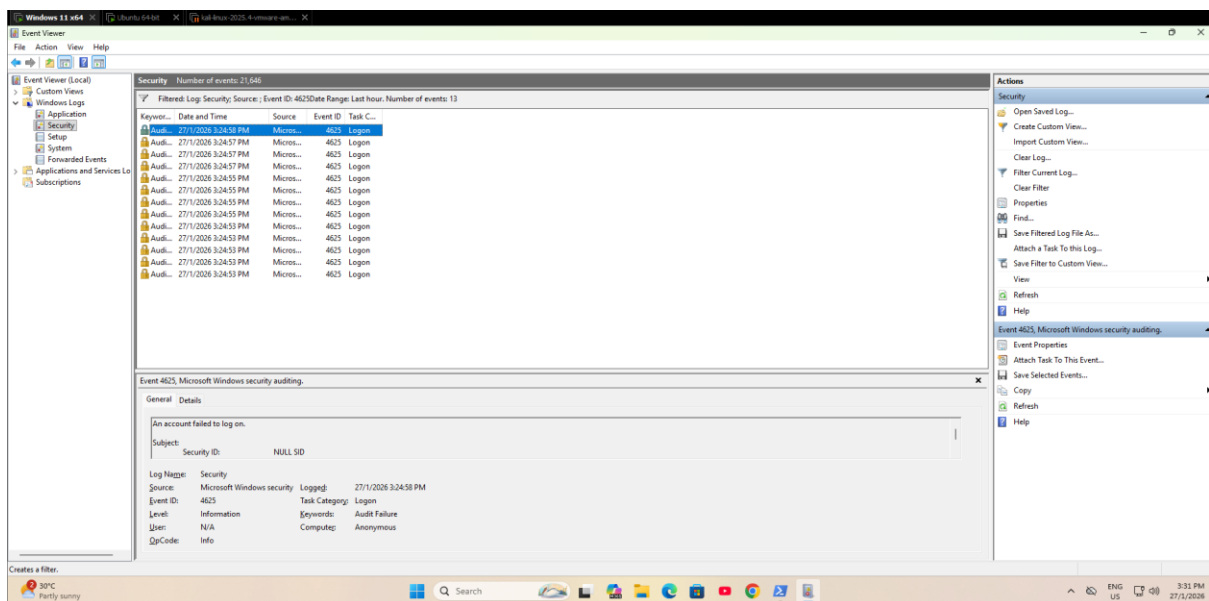
The Problem: Even after changing the port, the Splunk service on Ubuntu didn't have the "authority" to open new network sockets. It was running as a standard user without the right keys. **The Fix:** We used the --run-as-root flag and sudo. This gave Splunk the administrative power to bypass Linux permission restrictions and successfully bind to the network hardware.

3. The "Ghost Process" (Windows Service Hang)

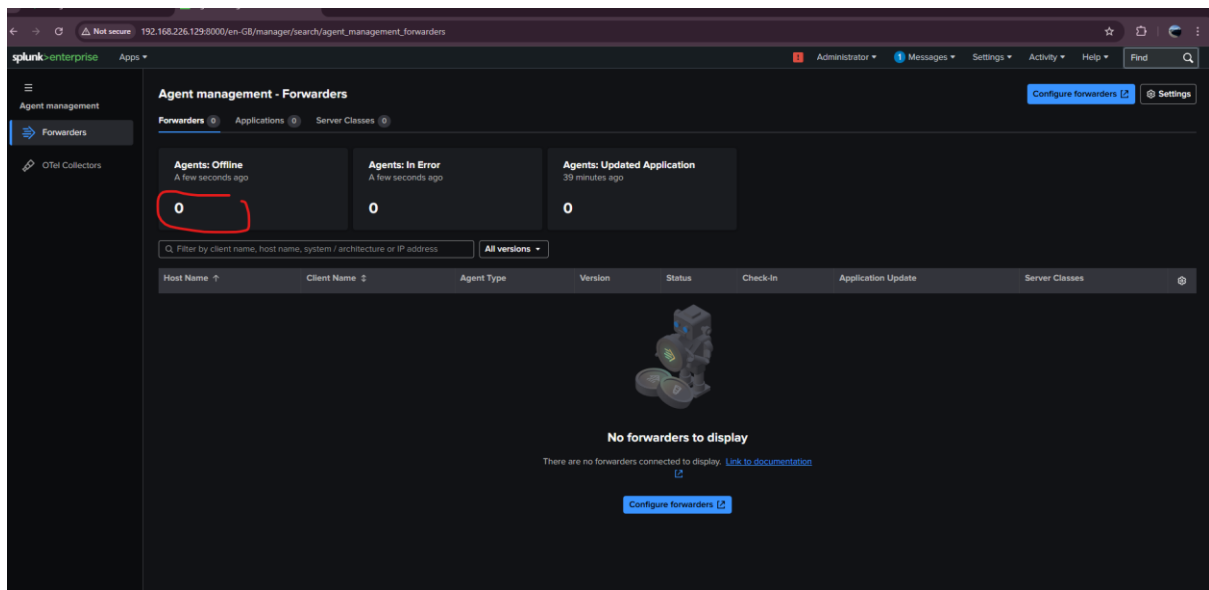
The Problem: On the Windows side, the Splunk Forwarder got "stuck" trying to talk to the old port. Even when we changed the config, the old process was still haunting the memory, refusing to let the new settings take effect. **The Fix:** We used taskkill to force the old process to die and then used Start-Service to give it a fresh, clean start with the new Port 9999 configuration.

```
kali@kali: ~  
Session Actions Edit View Help  
Nmap done: 1 IP address (1 host up) scanned in 0.54 seconds  
  
(kali@kali)~  
$ hydra -l Administrator -P /usr/share/wordlists/fasttrack.txt rdp://192.168.226.128 -t 4 -V  
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-01-27 02:24:42  
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.  
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 262 login tries (l:1/p:262), ~66 tries per task  
[DATA] attacking rdp://192.168.226.128:3389/  
[ATTEMPT] target 192.168.226.128 - login "Administrator" - pass "Spring2017" - 1 of 262 [child 0] (0/0)  
[ATTEMPT] target 192.168.226.128 - login "Administrator" - pass "Spring2021" - 2 of 262 [child 1] (0/0)  
[ATTEMPT] target 192.168.226.128 - login "Administrator" - pass "spring2021" - 3 of 262 [child 2] (0/0)  
[ATTEMPT] target 192.168.226.128 - login "Administrator" - pass "Summer2021" - 4 of 262 [child 3] (0/0)  
[ATTEMPT] target 192.168.226.128 - login "Administrator" - pass "summer2021" - 5 of 262 [child 0] (0/0)  
[ATTEMPT] target 192.168.226.128 - login "Administrator" - pass "Autumn2021" - 6 of 262 [child 1] (0/0)  
[ATTEMPT] target 192.168.226.128 - login "Administrator" - pass "autumn2021" - 7 of 262 [child 3] (0/0)  
[ATTEMPT] target 192.168.226.128 - login "Administrator" - pass "Fall2021" - 8 of 262 [child 2] (0/0)  
[ATTEMPT] target 192.168.226.128 - login "Administrator" - pass "fall2021" - 9 of 262 [child 0] (0/0)  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] freerdp: The connection failed to establish.  
[ATTEMPT] target 192.168.226.128 - login "Administrator" - pass "Winter2021" - 10 of 262 [child 3] (0/0)  
[RE-ATTEMPT] target 192.168.226.128 - login "Administrator" - pass "Autumn2021" - 11 of 262 [child 1] (0/0)
```

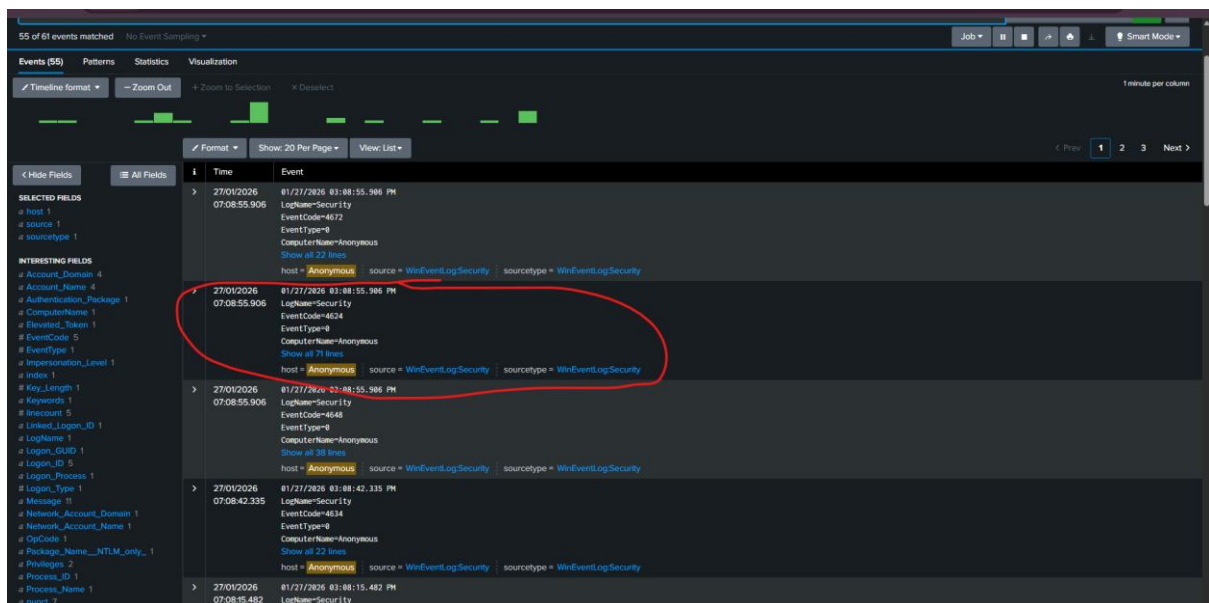
Executing a Brute-Force attack simulation using Hydra on Kali Linux. By targeting the RDP service on the Windows 11 endpoint, I am generating 'Audit Failure' events. This phase demonstrates the 'Detection' capability of the SIEM, as these high-frequency login failures will be captured and indexed by Splunk for alerting.



Windows Security Log viewing Event ID 4625. This confirms the local system is recording the unauthorized RDP connection attempts triggered by the Hydra tool



This is a common Splunk quirk. The "Agent Management" page often stays empty even when data is flowing perfectly. To see your actual data, you must use the **Search** bar.



Verification of successful log ingestion. The Splunk Search interface displays a count of 67 Security events received from the remote Windows 11 endpoint ('Anonymous'). This confirms that the custom port configuration (9999) and the Universal Forwarder are operational, providing the visibility necessary for incident detection.

Conclusion

The project successfully established a fully functional distributed SIEM architecture capable of collecting and analyzing security events from remote endpoints. By integrating Windows event logs with Splunk Enterprise and validating visibility through attack simulations, the environment demonstrated the core capabilities required for enterprise security monitoring.

This project strengthened skills in SIEM deployment, log ingestion, troubleshooting, network administration, and security operations while providing a solid foundation for advanced detection engineering and threat hunting activities.